

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

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Semester: Summer 2025-2026

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Date: July 25, 2026

Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. Which of the following is the primary objective of software project estimation?

- A. To determine programming language
- B. To predict effort, cost, and schedule
- C. To increase source code size
- D. To reduce documentation

Answer: B

2. Which factor most commonly contributes to software project failure?

- A. Excessive testing
- B. Poor communication among stakeholders
- C. High processor speed
- D. Frequent software updates

Answer: B

3. Which document formally defines project scope, objectives, assumptions, and constraints?

- A. User Manual
- B. Software Project Charter
- C. Test Report
- D. Installation Guide

Answer: B

4. Which quality attribute measures how easily software can be transferred to another platform?

- A. Maintainability
- B. Reliability
- C. Portability
- D. Availability

Answer: C

5. Which software engineering activity transforms requirements into architectural components?

- A. Coding
- B. System Design
- C. Maintenance
- D. Testing

Answer: B

6. Which software metric measures the number of executable statements in a program?

- A. Function Points
- B. Source Lines of Code (SLOC)
- C. Cyclomatic Complexity
- D. Defect Density

Answer: B

7. Which testing technique evaluates software without executing the program?

- A. Dynamic Testing
- B. Static Testing
- C. Stress Testing
- D. Recovery Testing

Answer: B

8. Which of the following is NOT a functional requirement?

- A. User login
- B. Report generation
- C. Response time below two seconds
- D. Password reset

Answer: C

9. Which software engineering activity identifies possible project risks before development starts?

- A. Risk Analysis
- B. Unit Testing
- C. Refactoring
- D. Deployment

Answer: A

10. In Agile projects, who is responsible for prioritizing business value?

- A. Scrum Master
- B. Product Owner
- C. Software Tester
- D. Development Manager

Answer: B

11. Which XP practice allows every programmer to modify any part of the source code?

- A. Pair Programming
- B. Collective Code Ownership
- C. Refactoring
- D. Continuous Testing

Answer: B

12. Which Scrum event usually lasts about 15 minutes?

- A. Sprint Planning
- B. Sprint Review

- C. Daily Scrum
- D. Sprint Retrospective

Answer: C

13. Which DSDM principle emphasizes building the correct solution rather than simply completing activities?

- A. Focus on Business Need
- B. Never Demonstrate Control
- C. Ignore User Feedback
- D. Delay Testing

Answer: A

14. Which FDD practice encourages regular visibility of project progress?

- A. Progress Reporting
- B. Pair Programming
- C. Continuous Deployment
- D. Sprint Retrospective

Answer: A

15. Which software engineering principle recommends dividing a complex system into smaller manageable modules?

- A. Modularity
- B. Encapsulation
- C. Serialization
- D. Virtualization

Answer: A

16. Which software document describes how software should be tested?

- A. Test Plan
- B. User Guide
- C. Maintenance Manual
- D. Requirement Traceability Matrix

Answer: A

17. Which software quality factor measures the ability of software to continue operating without failure?

- A. Efficiency
- B. Reliability
- C. Portability
- D. Reusability

Answer: B

18. Which UML diagram is primarily used to model interactions between users and the system?

- A. Class Diagram
- B. Sequence Diagram
- C. Use Case Diagram
- D. Activity Diagram

Answer: C

19. Which maintenance activity improves software readability and internal structure without changing external behavior?

- A. Refactoring
- B. Deployment
- C. Verification
- D. Requirement Elicitation

Answer: A

20. Which software engineering concept ensures that every requirement can be traced to its corresponding design, implementation, and test case?

- A. Requirement Traceability
- B. Continuous Integration
- C. Pair Programming
- D. Software Reuse

Answer: A